

A new method for preparing anhydride grafted polypropylene with high grafting ratio by combining microwave reaction and reactive extrusion

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Research Article

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Abstract

Maleic anhydride grafted polypropylene (PP-g-MAH) is an effective compatibilizer in the preparation of polypropylene composites. Its grafting ratio plays an important role in the dispersion of filler or blend in polypropylene (PP) matrix resin, but the grafting ratio of PP-g-MAH prepared by reactive extrusion is low. A method for preparing PP-g-MAH with high grafting ratio and low odor was proposed by combining microwave reaction and melt extrusion. The grafting ratio of the PP-g-MAH products obtained by this method without the assistance of comonomer reached 1.95%. The results of melt index and rheology showed that the crosslinking occurred during the preparation process, thus avoiding the large reduction of molecular weight of PP. Meanwhile, the problem of odor was avoided because no comonomer was added during the process. The application results showed that PP-g-MAH prepared by this method improves the compatibility of glass fiber (GF) and PP, thus improving the mechanical properties of polypropylene/glass fiber (PP/GF) composites. This work explored a novel way to prepare PP-g-MAH with high grafting ratio.

Introduction

Maleic anhydride grafted polypropylene (PP-g-MAH) is a commonly used compatibilizer in the production of polypropylene composites. The preparation method affects the grafting ratio of PP-g-MAH, thereby affecting its application performance. Its industrial preparation methods mainly include reactive extrusion, solution grafting, and suspension grafting [1]. Of these, reactive extrusion is the reactive extrusion of PP, maleic anhydride (MAH), and initiator that are uniformly mixed and put into the extruder, which has high efficiency [2, 3], yet suffers from a low grafting ratio of PP-g-MAH. To solve the problem of low grafting ratio, electron-donating monomers such as styrene (St), α -methylstyrene and scCO_2 are added in the preparation of MAH [4–8]. However, they produce a pungent odor that limits the application of PP-g-MAH [9, 10]. Furthermore, the reactive extrusion process causes extreme degradation of PP, directly impacting its performance [2, 11]. In recent years, the demands for PP have necessitated the development of a highly effective method for producing PP-g-MAH. Therefore, it is of great significance to develop a highly efficient PP-g-MAH preparation method with high grafting ratio without comonomers.

The microwave reaction has been widely used in polymer modification because of its fast reaction rate and low energy consumption. In recent years, it has been used in the preparation of PP-g-MAH. In this method, PP, MAH and initiator are mixed first, and then the polar components in the mixture are selectively heated by a microwave reactor for grafting reaction, and solid PP is not affected by microwave and will not degrade [12–15]. The grafting reaction occurs at the surface PP particles, resulting in low grafting efficiency. Some researchers used swelling method to solve the problem. Xylene is usually used as swelling agent [16]. During the swelling process, the swelling agent with MAH and initiator diffused into the PP particles. Then the wet mixture was irradiated by microwave, the reaction also occurred in the PP particles, increasing the reaction interface. Although the method can increase the grafting ratio of PP-g-MAH, it requires large amount of solvent and increased energy consumption of reaction. Up to now, there is no simple, effective method to prepare PP-g-MAH with a high grafting ratio.

In this paper, a novel preparation method for PP-g-MAH combining microwave reaction and reactive extrusion was proposed. The effects of BIPB, DCP and BPO as initiators on grafting ratio of PP-g-MAH were initially investigated, and the most suitable initiator system for microwave reaction was selected. Subsequently, the influence of combining microwave reaction and reactive extrusion on the grafting ratio of PP-g-MAH was explored. Moreover, the effect of PP-g-MAH preparation by this method on the mechanical properties of polypropylene/glass fiber (PP/GF) composites was examined, and the compatibilization mechanism was further discussed.

Materials and Methods

Materials

Polypropylene (PP, brand 045) was purchased from Zhejiang Hongji Petrochemical Co., Ltd. Maleic anhydride, with purity of 99%, was purchased from Shanghai Macklin Biochemical Co., Ltd. Benzoyl peroxide (BPO), Bis(tert-butyldioxyisopropyl) benzene (BIPB) and dicumyl peroxide (DCP) are 99% were all acquired from China National Pharmaceutical Co., Ltd, all with a purity of 99%. Xylene, ethanol, phenolphthalein, potassium hydrogen phthalate and other test reagents were additionally obtained from Shanghai Titan Technology Co., Ltd. PP-g-MAH (brand CMG-5701, grafting ratio 0.8 ~ 1.0%) was purchased from Jiayirong Co., Ltd. Glass fiber (GF, brand 508A), was purchased from Jushi Group Co., Ltd.

Preparation of PP-g-MAH by microwave reaction

Based on the selective heating property of microwave, PP-g-MAH powder was prepared by means of a solid-phase microwave reaction. PP, MAH and the initiator were mixed in a high-speed mixer for 5 minutes. Subsequently, they were placed in a 300watt microwave reactor for a 12-minute microwave grafting reaction. Then, the product was washed in hot ethanol, dried and stored for subsequent determination.

Preparation of PP-g-MAH by combining microwave reaction and reactive extrusion

PP-g-MAH was prepared by combining microwave reaction and reactive extrusion. Firstly, PP, MAH and initiator were mixed by a mixer. Then 300watt microwave was used to irradiate the mixture for 12 minutes for grafting reaction. Subsequently, new initiator was added and the mixture was immediately placed into a twin-screw extruder (HT-35), and the extruder was set to a screw speed of 150 rpm. The temperature of each zone of twin-screw extruder was shown in Table 1.

Table 1
Temperature of each zone in the extruder

Zone	Zone1	Zone2	Zone3	Zone4	Zone5	Zone6	Zone7	Zone8	Die
Temperature (°C)	140	160	190	190	180	180	180	180	180

Purification of PP-g-MAH

PP-g-MAH was accurately weighed 5-10g and dried in vacuum at 70 °C for 8h. After drying, it was put into Soxhlet extractor. The unreacted monomer in PP-g-MAH was extracted with acetone as solvent for 24 hours. After extraction, the PP-g-MAH was vacuum dried at 70 °C for 12h to remove acetone residue.

Application performance of PP-g-MAH

PP/GF composites were prepared by extrusion. PP was blended with 20wt% GF and 2wt% PP-g-MAH according to HG20539-1992. The mixture was extruded with a twin-screw extruder (HT-35) at a screw speed of 150 rpm. The process conditions of the extruder were shown in Table 1. Injection molding machine (EM120-V) is used to inject standard test splines. The process conditions of injection molding machine were shown in Table 2. The samples were stored in a constant temperature oven (25 °C) for 48h and then tested and characterized. The performance of PP-g-MAH was compared with that of CMG5701.

Table 2
The injection conditions

Injection temperature (°C)			Injection pressure (MPa)	Injection time (s)	Holding time (s)
Zone1	Zone2	Zone3			
210	210	210	45	15	10

Determination of PP-g-MAH grafting ratio

KOH-ethanol solution with a concentration of about 0.05M was prepared and its concentration was calibrated with potassium hydrogen phthalate. 0.5g purified PP-g-MAH was accurately weighed and put into beaker. After about 50mL xylene was added, beaker was heated with a magnetic stirrer, and PP-g-MAH was uniformly dissolved. Phenolphthalein reagent was added to the solution, and then the solution was titrated to red with KOH-ethanol solution. The grafting ratio was calculated with the following equation:

$$G = \frac{98.06 (V_2 - V_1) C}{2000M} \times 100\%$$

1

where G is the grafting ratio (%) of PP-g-MAH. C is concentration (mol/L) of potassium hydroxide-ethanol solution used in titration. V_2 (ml) and V_1 (ml) are the volumes of hydroxide-ethanol solution before and

after titration, respectively. M (g) is mass of titrated sample. The grafting ratio of each sample will be titrated three times to obtain the average value.

Characterizations

Fourier transform infrared spectrometer (FTIR, Nicolet IS10, Thermo Fisher) was used to characterized chemical structure of PP-g-MAH. First, a KBr disc was prepared by pressing machine. After heating the disc and PP-g-MAH, PP-g-MAH was coated on the disc. Data were collected by infrared scanning for 32 times. The scanning range is $400\text{--}4000\text{ cm}^{-1}$, and the resolution is 4 cm^{-1} .

The melt indexer (Goettfer, MI-4) was used to measure the flow performance of the PP-g-MAH. The melt flow rate (MFR) was tested at a temperature of 230°C and a load of 2.16 kg.

The rotational rheometer (MARS + 60, Thermo Fisher) was used to measure the rheological behavior of pure PP and PP-g-MAH. The geometric diameter of the parallel plates was 25 mm and the gap was 1 mm. Nitrogen atmosphere was used to avoid thermal degradation of samples. In the linear viscoelastic range of $0.1 \sim 628\text{ rad/s}$, the functional relationship of complex viscosity and storage modulus with angular frequency is showed.

Scanning electron microscope (SEM; Nova Nano 450, FEI) was used to observe fracture surface of PP/GF composites.

Gel Permeation Chromatograph (GPC, Waters China Limited) was used to determine the molecular weight of pure PP and PP-g-MAH. The mobile phase is trichlorobenzene, the test temperature is 150°C , the flow rate is 1.0 mL/min , and polystyrene is the standard sample.

Universal testing machine (MTS Industrial Systems, China) was used to test the mechanical properties of materials. The tensile tests were performed according to GB/T 1040–2018. The flexural tests were performed according to GB/T 9341 – 2008. Izod impact testing machine (China Chengde Kecheng Testing Machine Co., Ltd.) was used to test the impact performance of materials according to GB/T 1043–2008.

Results and Discussion

The effect of the initiator on the grafting ratio of PP-g-MAH in the microwave reaction process

Initiators play an important role in grafting reactions, thus the effects of three initiators, BIPB, BPO, and DCP, on grafting ratio were studied. The three initiators were mainly used in various grafting reactions.

The effect of the concentration of three initiators on the grafting ratio of PP-g-MAH was characterized by FTIR. The results are presented in Fig. 1.

Figure 1 showed the absorption spectra of pure PP and PP-g-MAH in the range of $2000\text{--}1000\text{ cm}^{-1}$. It can be seen that PP-g-MAH had new absorption peaks at 1710 and 1780 cm^{-1} compared to pure PP,

which can be attributed to the carbonyl group of MAH, indicating that MAH had been successfully grafted onto PP (Fig. 2). The grafting ratio of PP-g-MAH can be preliminarily determined by the strength of the peak [2].

The grafting ratio of PP-g-MAH prepared with BIPB and DCP as initiators first increased and then decreased with the increase of initiator concentration. The grafting ratio of PP-g-MAH prepared with BPO as initiator increased with the increase of concentration of initiator. The results indicated that the grafting efficiency of MAH was high when the concentration of BIPB and DCP was low, while that of MAH was low when the concentration of BPO was low. FTIR is difficult to quantitatively analyze the grafting ratio, so titration method was used. The results were shown in Fig. 3.

From Fig. 3, it was found that the grafting ratio of PP-g-MAH first increased and then decreased gradually with the increase of the concentration of BIPB and DCP. When the concentration of BIPB was 0.2%, the grafting ratio of PP-g-MAH was the highest, 1.21%. When the concentration of DCP was 0.4%, the grafting ratio of PP-g-MAH was 1.05%. The grafting ratio of PP-g-MAH gradually increased with the increase of BPO concentration. When the concentration of BPO was 1%, the grafting ratio of PP-g-MAH was 1.25%, and when the concentration was 0.2%, it had little effect on grafting ratio, so its ability to initiate grafting reaction was weak.

The microwave grafting reaction was a continuous heating process, and the temperature of the system gradually increased. The literature showed that the microwave reaction temperature remained around 206 °C within 10 minutes [12], such temperature changes make the temperature sensitivity of the initiator directly affect its initiation efficiency. Both BIPB and DCP are initiators with low temperature sensitivity, and their half time period at 180 °C is about 1 minute. It can be inferred that at the beginning of the microwave reaction, the decomposition rate of the two initiators was slow due to the low temperature of the system, and the reaction rate was extremely low at this time. The decomposition rate of the initiator increased rapidly with increasing temperature. When the concentration of BIPB was higher than 0.6wt%, it can be observed that the color of PP-g-MAH turned yellow, which was due to the self-polymerization of MAH [17]. Therefore, the rapid decomposition of high concentration initiators led to a large number of free radicals that can cause side reactions, so the grafting ratio of PP-g-MAH prepared with BIPB at low concentration was high. DCP and BIPB have the similar temperature-sensitive characteristics. The temperature sensitivity of DCP is slightly higher than that of BIPB, and the ability to initiate grafting is slightly poor. The decomposition rate of BPO at low temperatures was higher than that of BIPB and DCP. In the microwave reaction process, BPO started to initiate the reaction at a lower temperature, resulting in premature consumption. The grafting ratio of PP-g-MAH increased with the increase of the concentration of BPO, because more initiators can prolong the reaction time.

Based on the experimental results, although three initiators could initiate grafting reaction irradiated by microwave, the optimum concentration of BIPB was the lowest. Therefore, BIPB is the suitable initiator for the microwave grafting reaction.

The degradation of PP-g-MAH prepared by microwave reaction using BIPB as initiator

The degradation of PP caused by grafting reaction reduces the mechanical property of PP-g-MAH, thus affecting its application range. The degree of degradation of PP-g-MAH can be characterized by MFR, molecular weight and rheological properties of the melt.

The results showed that the MFR of pure PP was 3.97 g/10min. After the grafting reaction, the MFR of PP-g-MAH was slightly increased to 6.99 g/10min. The results of GPC showed that the M_w of pure PP and PP-g-MAH prepared with BIPB as initiator was 360900 g/mol and 299500 g/mol, respectively. It indicated that when BIPB was used as the initiator, the degradation of PP was not serious because its concentration was low, and the reaction only occurred at the surface of PP particles.

The rheological properties of PP-g-MAH were further characterized to investigate its degradation. Figure 4 (a and b) showed the curves of complex viscosity (η^*) and storage modulus (G') as a function of angular frequency (ω) for pure PP and PP-g-MAH melts at 180°C. Figure 4 (c and d) showed the intersection of G' - ω curve and G'' - ω curve for pure PP and PP-g-MAH.

From Fig. 4 (a), it can be observed that at the same frequency, the composite viscosity of PP-g-MAH was lower than that of pure PP, indicating that BIPB caused degradation of PP during the preparation of PP-g-MAH. From Fig. 4 (b), it can be observed that the slope of G' - ω curve of PP-g-MAH at low-frequency region was lower than that of pure PP, indicating that the relaxation time of PP-g-MAH was longer [18]. Given that PP-g-MAH is a non-filled system, the extension of the relaxation time means that the internal cross-linking occurred [19, 20]. As shown in Fig. 4 (c) and (d), the frequency of the intersection of the PP-g-MAH curve was lower than that of pure PP. According to Doi-Edwards theory, the reciprocal of the frequency at the intersection of the G'' - ω curve and the G' - ω curve was considered to be the polymer relaxation time, and the shift of the intersection to the low frequency region indicated the presence of cross-linking in the system [21]. Cross-linking compensates for the decrease in molecular weight caused by the decomposition of PP caused by the initiator, thereby improving the apparent melt viscosity of PP-g-MAH. Therefore, BIPB is the suitable initiator for preparation of PP-g-MAH.

Preparation of PP-g-MAH by combining microwave reaction and reactive extrusion

In industrial application, PP-g-MAH with higher grafting ratio was always required. However, it is difficult to increase the grafting ratio of PP-g-MAH by using microwave method alone. A novel preparation method for PP-g-MAH combining microwave reaction and reactive extrusion was proposed. According to the previous study, the concentration of MAH and BIPB was selected as 5% and 0.2% respectively. Since the combination process involves two stages, 0.1wt% BIPB was added to the mixture during the reactive extrusion to compensate for the loss of initiator at the first stage. The results of grafting ratio of PP-g-MAH were shown in Fig. 5.

The results showed that the grafting ratio of PP-g-MAH prepared by the combined process reach to 1.95%, while those prepared by the reactive extrusion method and microwave method were 0.65% and 1.21%, respectively. It can be seen that the grafting ratio of PP-g-MAH prepared by the combination process was almost the sum of those of the products prepared by two separate methods. This indicated that the microwave reaction will not affect the subsequent reactive extrusion process during the preparation of PP-g-MAH. The grafting ratio of PP-g-MAH prepared by reactive extrusion method alone was low, which was due to the low grafting activity and serious self-polymerization of MAH. The literature showed that, the reactive extrusion can obtain high grafting efficiency only with the aid of other comonomers such as styrene [22–24], however styrene often causes odor problems. Therefore, the combined process of microwave reaction and reactive extrusion is a simple and effective method for preparing odorless PP-g-MAH with high grafting ratio.

The color of PP-g-MAH was investigated because it affected its application. Figure 6 showed the color of pure PP and PP-g-MAH prepared by these methods. It can be seen that the color of pure PP was white from Fig. 6 (a). The color of PP-g-MAH (Fig. 6 (b)) prepared by microwave was light yellow, whereas the color of PP-g-MAH (Fig. 6 (c)) prepared by reactive extrusion was darker due to the self-polymerization of MAH. The color of PP-g-MAH (Fig. 6 (d)) prepared by the combined process was lighter than PP-g-MAH prepared by reactive extrusion, because the utilization of MAH was improved [17].

Figure 7 showed the MFR of pure PP and PP-g-MAH prepared by these methods. It can be seen that the MFR of pure PP and PP-g-MAH prepared by reactive extrusion method was 3.5g/10min and 33.4 g/10min, respectively. The result indicated that PP degraded due to intense shearing by twin screws and the action of the initiator during extrusion. The MFR of PP g-MAH prepared by the combined process was 23.7 g/10min, which was lower than that of PP g-MAH prepared by reactive extrusion. This indicated that the degradation of PP g-MAH prepared by combined process was not severe. The combined method included two steps, microwave reaction and reactive extrusion. On the one hand, the degradation of PP-g-MAH prepared by microwave grafting in the first step was not severe because the MFR of PP-g-MAH prepared by this method was only 6.99 g/10min. On the other hand, the initiator BIPB was consumed in large quantities in the first step, which resulted to significant decrease of BIPB content during reactive extrusion. Therefore, the combined process of microwave reaction and reactive extrusion did not cause severe degradation of PP.

Application properties of PP-g-MAH

In order to investigate the performance of PP-g-MAH prepared by combined process, it was used in the preparation of PP/GF composites. In industry, GF was always used to enhance the mechanical properties of PP materials. Due to the poor compatibility of the two materials, GF easily pull-out from PP/GF composites. To solve this problem, the most common solution is to add PP-g-MAH to improve the adhesion between PP and GF. In order to evaluate the application performance of PP-g-MAH, the mechanical properties of PP/GF composites with PP-g-MAH were investigated. Commercial PP-g-MAH CMG5701 was used for comparison. The results were shown in Fig. 8.

The results indicated that the impact strength, tensile strength and flexural modulus of PP/GF composites without compatibilizers were slightly higher than those of pure PP. After adding compatibilizers, the impact strength, flexural modulus, and tensile strength of PP/GF composites increased by 39.3%, 53.4%, and 20.7% compared with the PP-g-MAH without compatibilizer. Compared with the PP/GF composites added with CMG5701, the mechanical properties of the PP/GF composites added with PP-g-MAH prepared by the combined process had certain advantages.

In order to further explore the mechanism of PP-g-MAH improving the compatibility of PP/GF composites, the fracture surfaces images of pure PP and PP/GF composites were observed by SEM. The results were shown in Fig. 9.

Figure 9 (a) displayed a smooth and flat cross section of pure PP. Figure 9 (b) showed the cross section of PP/GF composites without compatibilizer. It can be seen that there were many large holes in the cross section. The large holes were caused by pulling-out of the aggregated GF during the impact process. Figure 9 (c) and (d) showed cross section of PP/GF composites with PP-g-MAH. It can be seen that the distribution of GF becomes uniform, indicating that PP-g-MAH can promote the uniform dispersion of GF in PP/GF composites. Many broken GF can be observed on the cross section, which indicated that the adhesion between GF and PP can be improved due to the addition of PP-g-MAH. When the PP/GF composites with compatibilizer was subjected to impact, GF mainly broke rather than pull-out, which required more energy, thus effectively improving the impact strength of PP [23, 25, 26].

Conclusions

In order to prepare PP-g-MAH with high grafting ratio, a new combined preparation process of PP-g-MAH based on microwave reaction and reactive extrusion was proposed. The optimum initiator and concentration for the preparation of PP-g-MAH were studied. The results showed that BIPB with a concentration of 0.2% was the most suitable initiator system for this reaction. FTIR results showed that MAH was successfully grafted onto PP. The grafting ratio of PP-g-MAH prepared by this method reached 1.95%, which was significantly higher than PP-g-MAH prepared by microwave reaction or reactive extrusion alone. Due to the absence of other monomers during the preparation process, the PP-g-MAH was odorless. Finally, the properties of PP-g-MAH as compatibilizer were studied by investigating the mechanical properties of PP/GF composites with PP-g-MAH. Compared to the PP/GF composites without compatibilizer, the impact strength, flexural modulus and tensile strength of PP/GF composites with PP-g-MAH were increased by 39.3%, 53.4% and 20.7%, respectively. The results showed that the addition of compatibilizer improved the dispersibility of GF in PP matrix and increased the adhesion between PP and GF, which was further confirmed by SEM results. To sum up, PP-g-MAH prepared by microwave reaction and reactive extrusion offers a few advantages, including high grafting ratio, high molecular weight, low irritating odor, and excellent color. This work explored a novel way to prepare PP-g-MAH with high grafting ratio.

Declarations

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Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Figures

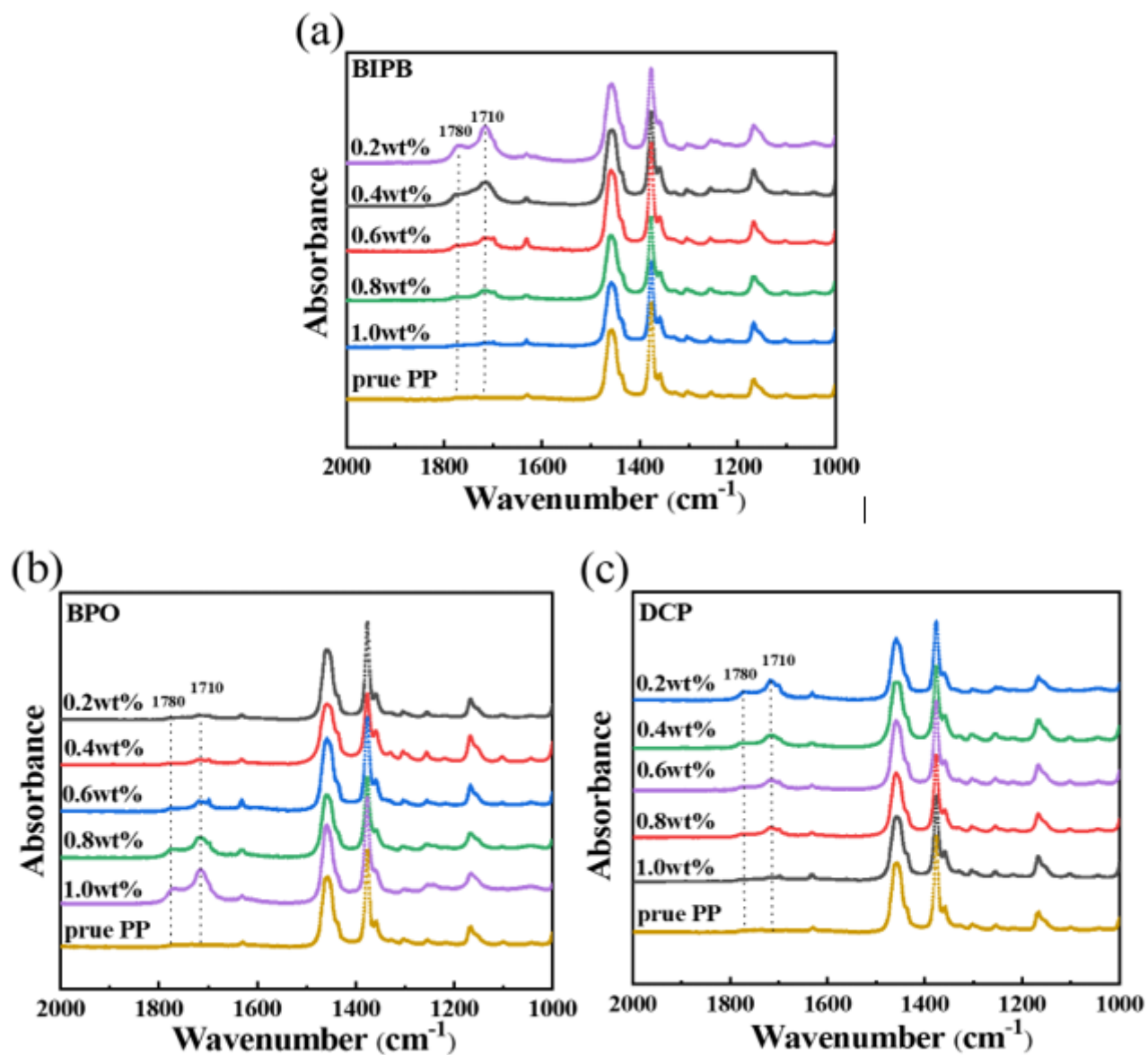


Figure 1

FTIR spectrum of PP-g-MAH prepared with (a) BIPB, (b) BPO, (c) DCP as initiator

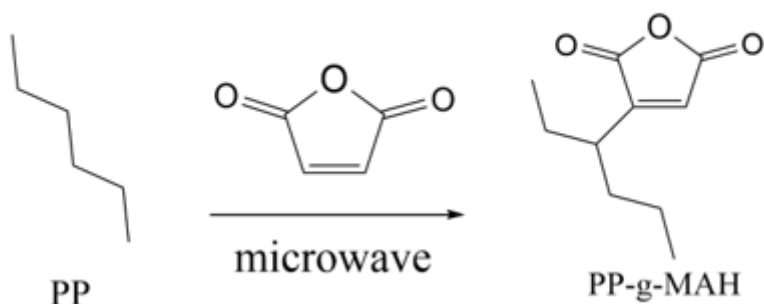


Figure 2

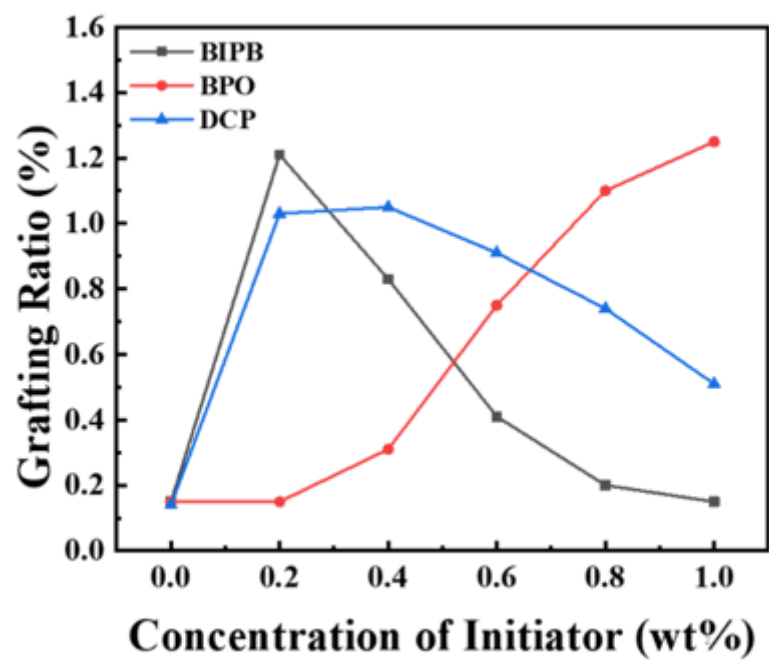


Figure 3

Effect of initiators on grafting ratio of PP-g-MAH

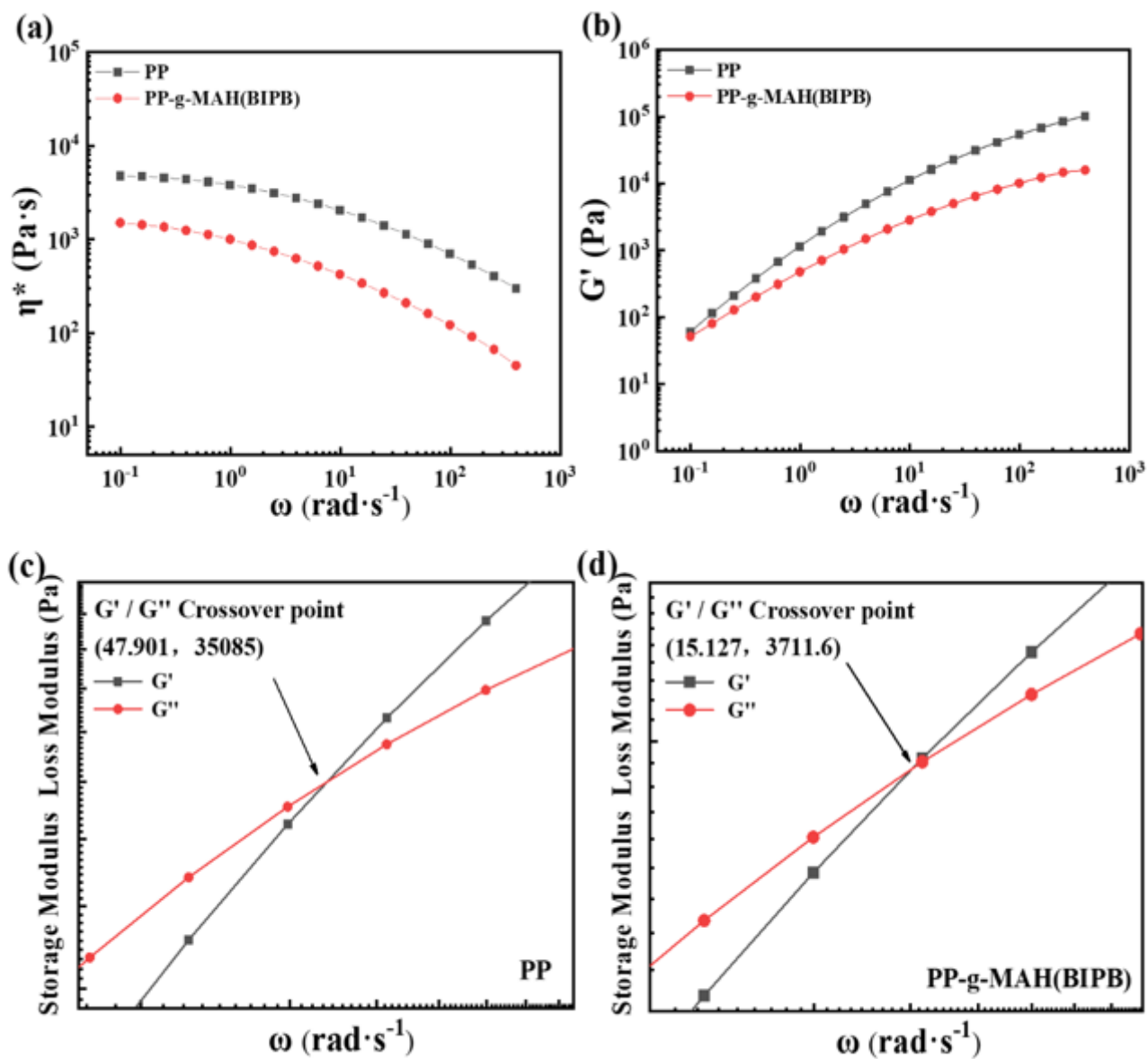


Figure 4

Complex viscosity (a) and storage modulus (b) of PP, PP-g-MAH blends as a function of angular frequency at 180 °C. Intersection of G' - ω curve and G'' - ω curve for pure PP (c) and PP-g-MAH (d)

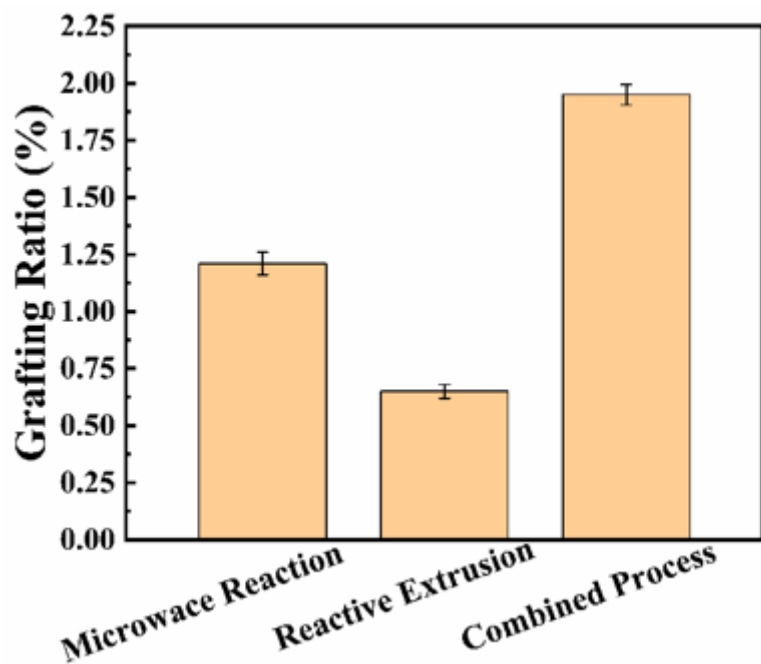


Figure 5

Grafting ratio of PP-g-MAH prepared by different processes

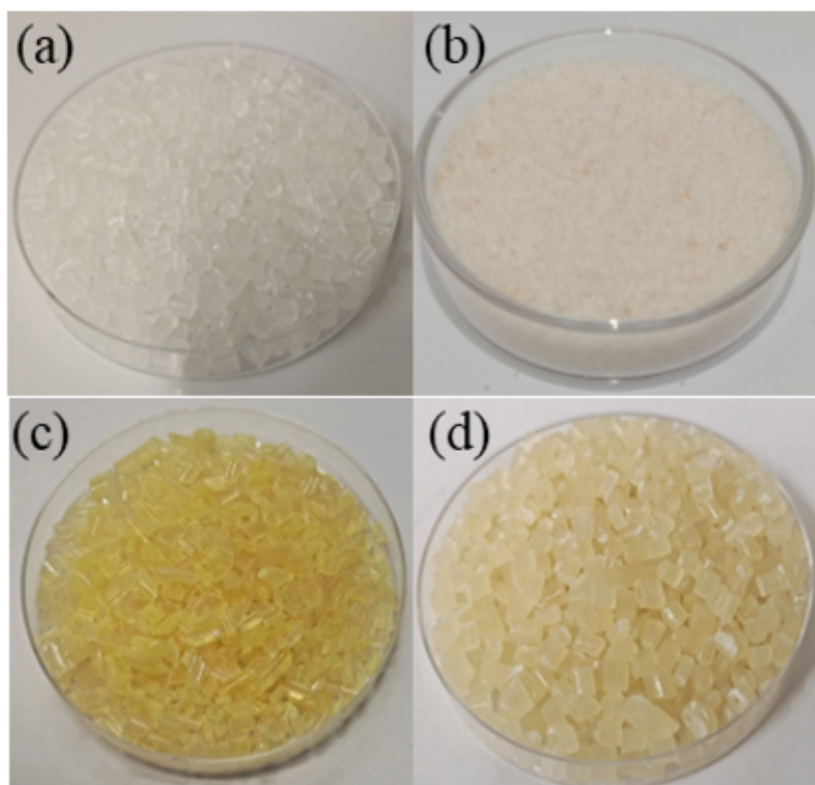


Figure 6

Color of pure PP (a), PP-g-MAH prepared by microwave reaction(b), reactive extrusion (c), combined process (d)

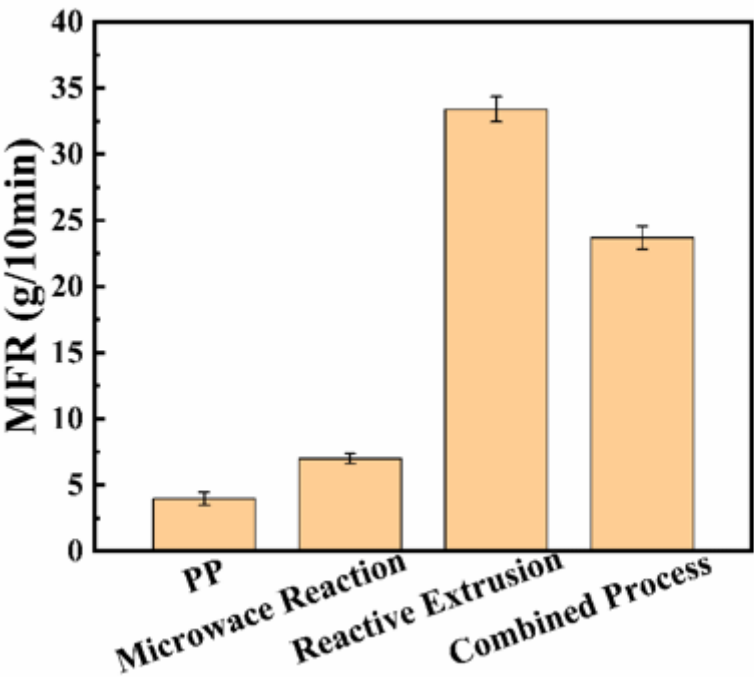


Figure 7
MFR of PP and PP-g-MAH prepared by different processes

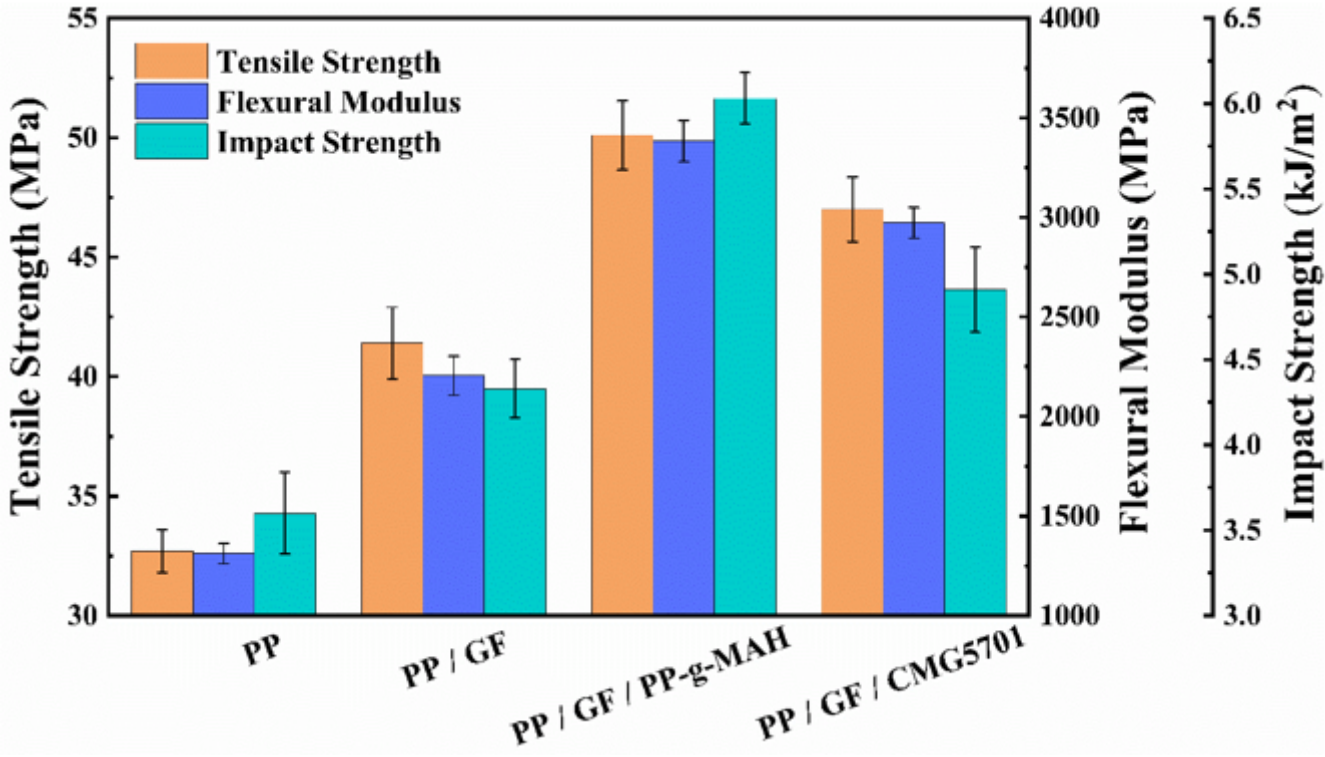


Figure 8

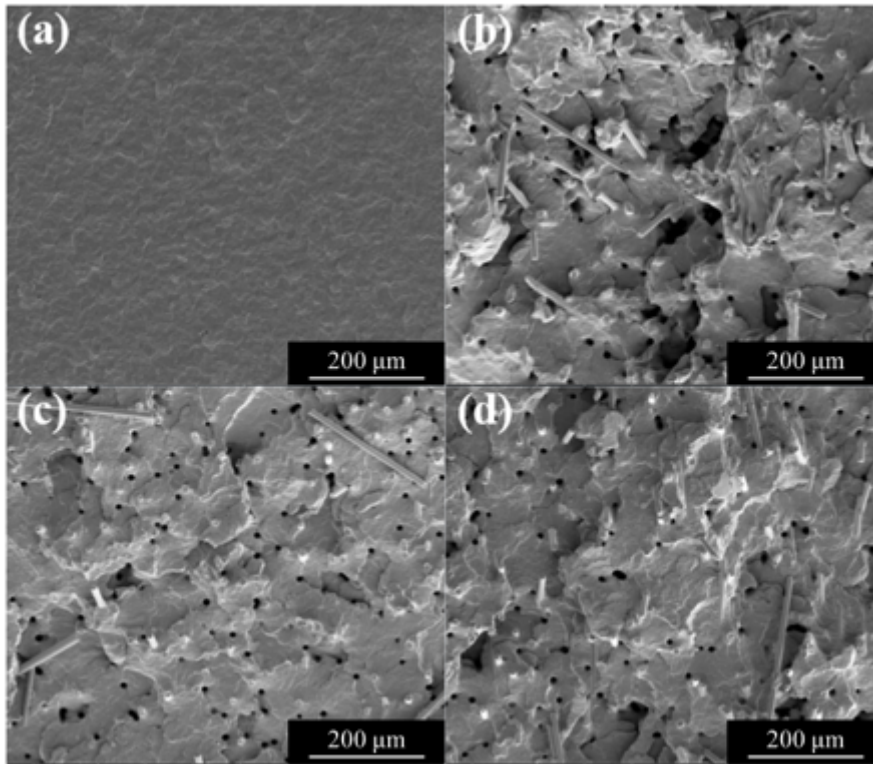


Figure 9

SEM image of the fracture surfaces of PP (a), PP/GF composites without compatibilizer (b) and PP/GF composites with PP-g-MAH prepared by combined process (c) and CMG5701 (d)